

RoboSumo:

Intro:

Robo Sumo is an exhilarating robotics competition where two robots battle to push each other out of a circular arena. This event challenges teams to design and build autonomous or remote-controlled robots within specific weight and size limits. It tests engineering, strategy, and control, with scoring based on successful pushes, tactical maneuvers, and compliance with competition rules.

Venue/Location: SCAS Annexure 2 (open air)

Arena Specifications:

- **Shape:** Circular ring (Dohyo)
- **Material:** Plywood
- **Dimensions:** 180cm diameter
- **Border:** White boundary line indicating the edge of the ring

Event Structure:

- The competition will follow a knockout or league-based format.
- Each match will last between 1-3 minutes per round.
- A robot wins if it successfully forces the opponent out of the arena, disables the opponent for 10 seconds, or wins through a referee decision in case of a deadlock.

Rules and Regulations Robot Specifications:

- **Weight Categories:**
- Middleweight: Up to 3 kg
- Heavyweight: Up to 5 kg
- **Size Constraints:**
- Robots must fit within 30 cm x 30 cm x 30 cm at the start of the match.
- **Power Source:**
- Battery-operated only (no combustion engines), only mounted onto the bot
- Maximum voltage: 24V.

- **Control System:**
- Robots should only be remotely controlled.
- Wireless communication must comply with event standards.
- **Weapons & Attachments:**
- No destructive weapons allowed.
- Pushing mechanisms are NOT allowed ,scoops are permitted.

Match Rules:

- Robots are placed inside the ring with a countdown before activation.
- Allowed actions include pushing, lifting, flipping, and defensive maneuvers.
- Prohibited actions include damaging opponent robots, using liquids, fire, explosives, or external power sources.

Ticket Requirement

- Team Individuals must register in advance and pay the registration fee before the deadline and must have the RoboYudh 25 Ticket for participating.
- Each team can consist of 1-4 members.
- Pre-match verification of weight, size, and safety compliance is mandatory.

Marking Scheme / Criteria Scoring Parameters:

- Pushing the opponent out of the arena: 10 points
- Flipping the opponent: 5 points
- Aggression: Up to 5 points
- Rule adherence: Up to 5 points

Victory Conditions:

- Knockout Tournament

Safety Guidelines Robot Safety:

- A power cutoff switch is mandatory.

Arena Safety:

- Only authorized personnel are allowed inside the arena. Participant Safety:
- No direct physical contact with robots during matches.

Appeals and Disputes

- Teams may request a review within 5 minutes of match completion.
- Referees and event judges will handle all disputes and their decisions are final.
- Video review (if available) may be used in case of disputes.

Disqualification Minor Penalties (Warnings issued):

- Deliberate stalling
- Exceeding time limits

Major Penalties (Disqualification possible):

- Intentional destruction
- Safety violations
- Use of prohibited weapons
- Interfering with other teams
- Failure to comply with the event timeline